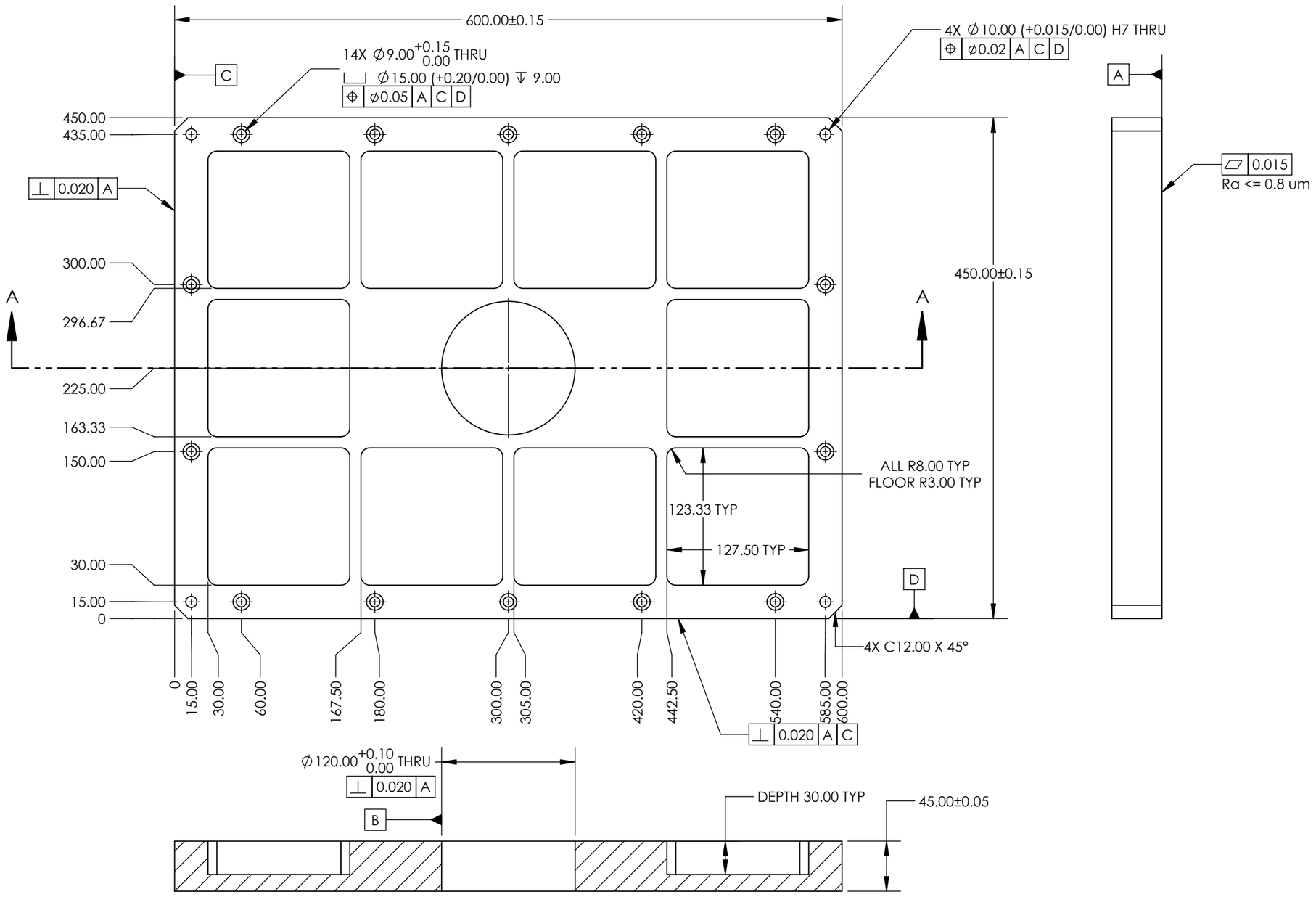
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-SEMI-ATE-BASEPLATE-8.1.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR ISO 2768-f (±0.30 mm FOR SPAN >400mm), ANGULAR ±0.1 DEG.
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM C-D-A).

2. MANUFACTURING & CNC DFM DISCIPLINE:
2.1 MATERIAL: FORGED AEROSPACE ALUMINUM ALLOY 7075-T651 (AMS 4045, PRE-STRETCHED PLATE, YIELD STRENGTH >= 505 MPa).
2.2 PROCESS ROUTE: 100% SUBTRACTIVE 3-AXIS CNC VERTICAL MILLING ONLY. ZERO UNDERCUTS, ZERO SLOPED HOLES.
2.3 CAVITY TOPOLOGY: ALL POCKET VERTICAL CORNERS CARRY R8.00 mm RADII (>= 1.3x CUTTING TOOL RADIUS) TO PREVENT CORNER TOOL CHATTER AND HIGH-SPEED FLUTE CHIPPING.
2.4 RIB STIFFNESS: INTERNAL WEBS MAINTAIN 10.00 mm SOLID WALL THICKNESS (ASPECT RATIO 3:1) RESISTING G-FORCE VIBRATION.

3. KINEMATIC RIGIDITY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (BOTTOM MOUNTING BASE): FLATNESS <= 0.015 mm ACROSS 600x450 mm ENVELOPE. SURFACE FINISH Ra <= 0.8 MICRONS.
3.2 SECONDARY DATUM [B] (CENTER CABLE APERTURE Ø120.00 THRU): PERPENDICULAR TO [A] WITHIN 0.020 mm.
3.3 4X DOWEL PIN ALIGNMENT BORES: NOMINAL SIZE Ø10.000 +0.015/0.000 (H7) THRU. TRUE POSITION WITHIN Ø0.02 mm RELATIVE TO [A], [C], [D]. PRE-MACHINING CAM OFFSET (+0.030 mm) APPLIED TO COMPENSATE 30 µm TYPE III ANODIZE.
3.4 14X BED MOUNTING FASTENERS: M8 COUNTERBORES CENTERED ON TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [A], [C], [D].

4. SURFACE ENGINEERING & CLEANROOM ASSURANCE:
4.1 100% DEBURRED UNDER MAGNIFICATION. ALL EXPOSED EDGES BROKEN C0.50 mm MAX.
4.2 SURFACE TREATMENT: HARD ANODIZE TYPE III (MIL-A-8625F, 30 µm THICKNESS, PTFE HYDROTHERMAL SEALING). COATING MICRO-HARDNESS >= 450 HV.
4.3 CLEANROOM QUALIFICATION: 100% BULK SUBTRACTIVE MACHINING TO ENSURE ZERO RESIDUAL POWDER PARTICULATE SHEDDING. FULLY CERTIFIED FOR SEMICONDUCTOR ISO CLASS 1 / FED STD 209E CLASS 1 ENVIRONMENTS.
4.4 PACKAGING: ULTRASONICALLY CLEANED WITH 18 MOhm·cm DEIONIZED WATER, SEALED IN DUAL-LAYER PE ANTI-STATIC CLEAN BAGS.



SECTION A-A

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ISO 2768-f (±0.30 mm for >400mm span), Angular ±0.1°	FINISH: Hard Anodize Type III (MIL-A-8625F, 30 µm, PTFE Hydrothermal Sealed)				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION: A
						Mode 1: 3-Axis CNC Straight Milling (Zero Undercuts, 100% Subtractive)
	NAME	SIGNATURE	DATE			TITLE: WAFER-LEVEL ATE TEST HANDLER HEAVY-DUTY GANTRY BASEPLATE
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Forged Aerospace Aluminum Alloy 7075-T651 (AMS 4045, Pre-Stretched Plate, Density 2.81 g/cm³)		DWG NO. W3-SEMI-ATE-BASEPLATE-8.1 / A3
				WEIGHT: ~17.85 kg		SCALE NTS / SHEET 1 OF 1